Inverse Hyperbolic Functions

Questions 1–17

Question 1. Given $y = \sinh x$, show that

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$$

Question 2. Given $y = \cosh x$, show that

$$\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$$

stating the values of x for which $\operatorname{arcosh} x$ exists.

Question 3. Given $y = \tanh x$, show that

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

stating the values of x for which $\operatorname{artanh} x$ exists.

Question 4. Write down the exact values of the following:

a) $\operatorname{arsinh} 1$
b) $\cosh^{-1} 2$
c) $\operatorname{artanh} \frac{1}{3}$
d) $\sinh^{-1}(-4)$
e) $\operatorname{arcosh} 9$
f) $\tanh^{-1}(-\frac{1}{2})$

Question 5. Solve the following equations, writing your answers in exact form:

a) $\sinh x = 0$

b) $\sinh x = 3$

c) $\sinh x = -6$

Question 6. Solve the following equations, writing your answers in exact form:

a) $\cosh x = 1$

b) $\cosh x = 6$

c) $\cosh x = \frac{3}{2}$

Question 7. Solve the following equations, writing your answers in exact form:

a) $\tanh x = 0$
b) $\tanh x = \frac{1}{4}$
c) $\tanh x = -\frac{5}{6}$

Question 8. Solve the following equations, writing your answers in exact form:

a) $\sinh 2x = 5$
b) $\cosh 3x = 8$
c) $5\tanh(3x - 1) = 2$
d) $3\sinh(2 - \frac{1}{3}x) = 1$

Question 9.

a) Sketch and label the graphs of $y = \sinh x$ and $y = \cosh x$ on the same axes.

b) Prove that the graphs of $y = \sinh x$ and $y = \cosh x$ do not intersect.

Question 10.

a) Sketch and label the graphs of $y = \cosh x$ and $y = \tanh x$.

b) Given that $\cosh^2 x \equiv 1 + \sinh^2 x$, prove that the graphs of $y = \cosh x$ and $y = \tanh x$ do not intersect.

Question 11. Solve the following equations, writing your answers in exact form:

a) $\sinh^2 x + \sinh x - 12 = 0$

b) $2\cosh^2 x - \cosh x - 21 = 0$

c) $12\tanh^2 x + \tanh x - 1 = 0$

Question 12. Solve the equation $3\cosh^2 x + 10\cosh x - 25 = 0$ writing your answers in the form $\ln k$, where k is a rational number.

Question 13. Solve the equation $3\cosh^2 x - 17\cosh x - 56 = 0$ writing your answers in the form $\ln(a + b\sqrt{7})$, where a and b are integers.

Question 14. Solve the following equations:

a) $\operatorname{arsinh} x = \ln 2$
b) $\operatorname{arcosh} x = \ln 3$
c) $\operatorname{artanh} x = \ln 4$
d) $\operatorname{arsinh} 2x = 2\ln 3$
e) $\operatorname{arcosh} 3x = 3\ln 2$
f) $\operatorname{artanh} \frac{1}{2}x = 2\ln \frac{1}{2}$

Question 15. Solve the following equations:

a) $x = \sinh(\ln(3 - x))$

b) $\cosh(\ln(2x - 1)) = x + 2$

c) $x - 1 = \tanh(\ln\sqrt{x + 1})$

Question 16. Solve the following equations:

a) $\operatorname{artanh} x - \ln(1 + x) = \ln\frac{6}{5}$

b) $\ln 2 + \ln(1 + x) - \operatorname{artanh} x = \ln\frac{2}{9}$

c) $\operatorname{arsinh} x - \ln(1 + x) = \ln 2$

d) $\ln 3 + \ln(x - 1) - \operatorname{arcosh} x = 0$

Question 17. Solve the following equations:

a) $\operatorname{arsinh} x + \ln 2 = \operatorname{arcosh} x + \ln 3$

b) $\operatorname{arsinh} x - 2\operatorname{arcosh} x = 0$